

Landfill_emissions_r1.R

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Contents

Municipal_solid_waste	1
Index	4

Municipal_solid_waste	<i>Create gridded municipal solid waste methane emissions maps</i>
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Description

‘Municipal_solid_waste’ writes 2 netcdf files of gridded methane emissions from municipal landfills, as well as optional visuals and more easily read csv files

Usage

```
Municipal_solid_waste(  
  LMOP_file,  
  domain,  
  state_name_list,  
  output_directory,  
  inventory_year,  
  verbose,  
  GHGI_landfill_total,  
  plot_directory,  
  County_Tigerlines,  
  State_Tigerlines,  
  focus_city_tigerlines  
)
```

Arguments

LMOP_file	Character providing the full filepath to the landfill methane outreach program’s landfill-level only data excel file available at https://www.epa.gov/lmop/landfill-technical-d Only the LMOP_database tab is used and only the variables GHGRP_ID, latitude, and longitude are used. Columns represent separate variables and rows represent separate landfills. There is an example file in the package’s datasets folder that has been successfully used in this code available for reference.
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domain	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system
state_name_list	Character vector listing all states within the desired domain
output_directory	Character providing the full filepath to save processed data
inventory_year	Numeric indicating the desired year of data to use.
verbose	Logical indicating whether to save additional output. This includes 4 csv files, 2 providing summarized and 2 providing detailed information for all landfills within the domain, separated between LMOP and GHGRP. It also includes 2 plots of the gridded methane emissions on log scales, one for LMOP facilities and one for GHGRP facilities.
GHGI_landfill_total	Numeric. Pulled from config file. The total national emissions from municipal solid waste from the GHGI for the inventory_year in kilotons per year, equivalent to gigagrams per year. The GHGI is available at https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks and the value can be found in the table titled 'CH4 emissions from Landfills (kt)' as row 'MSW net CH4 Emissions'.
plot_directory	Character providing the full filepath to save figures. Only relevant if verbose = TRUE.
County_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
State_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
focus_city_tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if a focus city was set in main and verbose=TRUE.

Details

This function calculates and grids methane emissions from municipal landfills. It uses the Environmental Protection Agency's (EPA) Greenhouse Gas Reporting Program (GHGRP) emissions when available. It then calculates the difference between national GHGRP emissions and the EPA national Greenhouse Gas Inventory (GHGI) emissions (GHGI - GHGRP) and distributes this residual equally to all facilities in the EPA Landfill Methane Outreach Program (LMOP) that are not already accounted for by the GHGRP.

The necessary GHGRP data will be automatically downloaded.

The GHGRP includes only facilities that emit at least 25,000 metric tons of carbon dioxide equivalent while the GHGI is intended to capture all national emissions and LMOP is a voluntary program with location and other details, but no emissions information. GHGRP data is available starting in 2010 and generally is about 2 years behind present day, the GHGI is available starting in 1990 and is updated approximately in sync with the GHGRP, and the LMOP is updated more frequently, sometimes multiple times per year. All three datasets are annual. The GHGRP and LMOP are at the facility scale while the GHGI is national totals.

Index

Municipal_solid_waste, [1](#)